

Task 1: Data preparation and customer analytics

We need to present a strategic recommendation to Julia that is supported by data which she can then use for the upcoming category review. However, to do so, we need to analyse the data to understand the current purchasing trends and behaviours. The client is particularly interested in customer segments and their chip purchasing behaviour. Consider what metrics would help describe the customers' purchasing behaviour.

We have chosen to complete this task in R, however you will also find Python to be a useful tool in this piece of analytics. If you aren't familiar with R or Python we would recommend searching a few online courses to help get you started. We have also provided an R solution template if you want some assistance in getting through this Task. Whilst its possible to complete the task in Excel you may find the size of the data and the nature of the tasks is such that it is more difficult to complete in Excel.

To get started, download the resource csv data files below and begin performing high-level data checks such as:

- Creating and interpreting high-level summaries of the data
- Finding outliers and removing these (if applicable)
- Checking data formats and correcting (if applicable)

You will also want to derive extra features such as pack size and brand name from the data and define metrics of interest to enable you to draw insights on who spends on chips and what drives spends for each customer segment. Remember, our end goal is to form a strategy based on the findings to provide a clear recommendation to Julia the Category Manager so make sure your insights can have a commercial application.

Task 2: Experimentation and uplift testing

Julia has asked us to evaluate the performance of a store trial which was performed in stores 77, 86 and 88.

We have chosen to complete this task in R, however, you will also find Python to be a useful tool in this piece of analytics. We have also provided an R solution template if you want some assistance in getting through this Task.

To get started, use the QVI_data dataset below or your output from task 1 and consider the monthly sales experience of each store.

This can be broken down by:

- total sales revenue
- total number of customers
- average number of transactions per customer

Create a measure to compare different control stores to each of the trial stores to do this write a function to reduce having to re-do the analysis for each trial store. Consider using Pearson correlations or a metric such as a magnitude distance e.g. $1 - (\text{Observed distance} - \text{minimum distance}) / (\text{Maximum distance} - \text{minimum distance})$ as a measure.

Once you have selected your control stores, compare each trial and control pair during the trial period. You want to test if total sales are significantly different in the trial period and if so, check if the driver of change is more purchasing customers or more purchases per customers etc.

Task 3: Analytics and commercial application

With our project coming to an end its time for us to send a report to Julia, based on our analytics from the previous tasks. We want to provide her with insights and recommendations that she can use when developing the strategic plan for the next half year.

As best practice at Quantum, we like to use the “Pyramid Principles” framework when putting together a report for our clients. If you are not already familiar with this framework you can find quick introductions on by searching form them on the internet.

For this report, we need to include data visualisations, key callouts, insights as well as recommendations and/or next steps.

We recommend you use a tool like PowerPoint (or similar) to create your report, we have also provided you a template to help get you started.

One way to create your visualisations (graphs, charts etc) is to save your output files from task 1 & 2 and create generate your charts in Excel. You may find it helpful to create chart templates however that is not essential. We are looking for consistency here – if a series is blue in one chart it needs to remain blue throughout the presentation. Once you have created the charts in your Excel file, paste them into your PowerPoint in the order that suits best.